

## Title

Optimizing Photothermal Therapy for Acne: A Randomised Protocol Comparing Laser-Enhanced versus Ultrasound-Assisted Nanoparticle Delivery

## Abstract

Acne vulgaris treatment using photothermal approaches targets sebaceous structures with minimal systemic exposure. This randomised study will compare two delivery mechanisms for enhanced gold nanoparticles incorporated in lipid-based carriers to the skin of patients with moderate-to-severe acne. One group will receive fractional CO<sub>2</sub> laser microperforation of the epidermis prior to nanoparticle application, while the second group will undergo ultrasound-assisted transdermal delivery. Following nanoparticle deposition, all participants will receive thermal laser treatment to activate photothermal destruction of targeted follicular structures. Clinical assessments will document lesion counts across comedonal, inflammatory, and cystic categories, as well as photographic documentation of scarring. Skin biopsies will characterise histological changes including follicular damage and collagen remodelling. Blood and urine analyses will quantify systemic absorption of gold nanoparticles. Safety monitoring will track adverse reactions, tolerability scores, and post-treatment recovery. This study aims to clarify whether mechanical barrier disruption via fractional ablation enhances therapeutic delivery and efficacy compared to acoustic delivery methods.